



Amrita School of Computing, Chennai

Course Delivery Plan

Course Code : Course Name	23CSE304 : Embedded Systems	Program	B.Tech. – CSE
L - T - P - C	3 – 0 – 2 – 4	Semester / Year	V / III
Name(s) of the Faculty	Dr.S.Natarajan, Dr.J.V.Aravind	Pre-requisite	23CSE213 Computer Organization and Architecture
Course Mentor	Dr.S.Natarajan	Academic Year	2026-2027
Course Overview	This course will enable the students to understand the basics of embedded systems and to develop an embedded system using ARM architecture.		

Course Objectives		Course Outcomes	
1	Through this course, the students will learn how microprocessors/ controllers work, microcontroller architecture basics, and how electronic gadgets are designed, developed, and built as embedded systems.	CO1	Understand the fundamentals, functions, and architecture of embedded systems.
2	This course emphasizes on learning by doing and building solutions to real-world problems using embedded systems.	CO2	Develop applications making use of GPIOs and peripheral units using Embedded C.
		CO3	Develop applications for the usage of Timers, Interrupts, and Debugging of embedded applications.
		CO4	Design an embedded system using an ARM processor.

Syllabus

Unit 1

Basics of Embedded Systems – Definition, Characteristics, Architecture of Microprocessors: General definitions of computers, microprocessors, microcontrollers, and digital signal processors. Introduction to ARM Cortex M4 Microcontroller – Architecture – Registers – Reset, Memory. Operating modes – Addressing modes and operands. Instructions for memory access, logical, arithmetic, and shift operations. Introduction to Thumb ISA.

Unit 2

GPIO programming- GPIO Configuration, Interfacing IO devices and its type – LEDs, Switches, Buzzer, Seven Segment Display, LCD, Analog I/O interfacing - DAC and ADC, PWM, Real-Time Clock, I/O synchronization.

Unit 3

Interrupt structure - Vector interrupt table, Interrupt service routines – Edge triggered, Timer Periodic Interrupts, NVIC, SysTick timer Synchronous and asynchronous communication, UART interfacing. I2C & SPI-I2C-specification, Protocol configuration, I2C Peripherals. SPI Specification, Protocol configuration- Abstraction -- Moore and Mealy machines, quantitative and qualitative performance measurements. Testing and Debugging of embedded applications.

Textbook(s)

T1. Jonathan W. Valvano.“Embedded Systems: Introduction to ARM Cortex-M Microcontrollers”. CreateSpace Independent Publishing Platform , 2016.

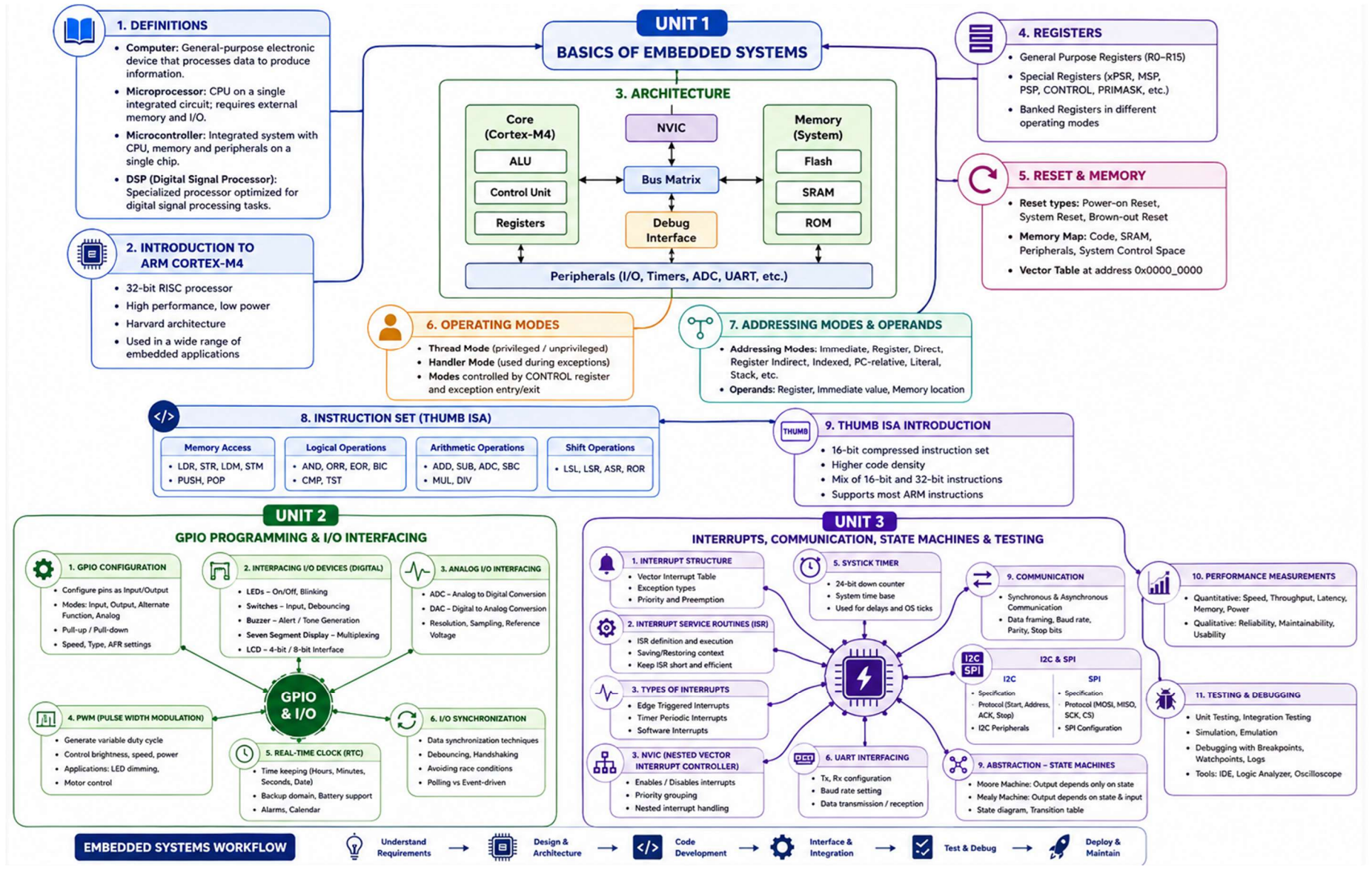
Reference(s)

R1. Furber SB. “ARM system-on-chip architecture”, Pearson Education; 2000.

R2. Ariel Lutenberg, Pablo Gomez, Eric Pernia. “A Beginner's Guide to Designing Embedded System Applications on Arm Cortex-M Microcontrollers”, ARM Education media, 2022.

R3. Joseph Yiu. “System-on-Chip Design with Arm Cortex-M Processors Reference Book”, ARM Education media, 2019.

Concept Map



Evaluation and Grading

Internal (70)				External (30)	Total
Components		Marks	Total Marks	End Semester = 30 (50 Marks; 2 hours exam)	Internal + External = 100
Mid Term Exam	Mid Term	20	20		
Continuous Assessment Theory	Quiz 1	5	10		
	Quiz 2	5			
Continuous Assessment Lab	Lab Activity Sheets	10	40		
	Capstone Project	20			
	Value Added Programme	5			
	Class Room Participation	5			
Programme Outcome (PO)					
PO1	Engineering Knowledge: To apply knowledge of mathematics, science, engineering fundamentals, problem solving skills, algorithmic analysis to solve complex engineering problems.				
PO2	Problem analysis: To analyze the problem by finding its domain and applying domain specific skills.				
PO3	Design/development of solutions: To understand the design issues of the product/software and develop effective solutions with appropriate consideration of public health and safety, cultural, societal, and environmental issues.				
PO4	Conduct investigations of complex problems: To find solutions of complex problems by conducting investigations applying suitable techniques.				
PO5	Modern tool usage: To adapt the usage of modern tools and recent software.				
PO6	The engineer and society: To contribute towards the society by understanding the impact of Engineering on global aspect.				
PO7	Environment and sustainability: To understand environment issues and design a sustainable system.				
PO8	Ethics: To understand and follow professional ethics.				
PO9	Individual and team work: To function effectively as an individual and as member or leader in diverse teams and interdisciplinary settings.				
PO10	Communication: To demonstrate effective communication at various levels.				
PO11	Project Management and finance: To apply the knowledge of Computer Engineering for development of projects, and its finance and management.				
PO12	Life-Long Learning: To keep in touch with current technologies and inculcate the practices of lifelong learning.				

PSO1	Ability to design and engineer, innovative, optimal and elegant computing solutions to interdisciplinary problems using standard practices, tools and technologies.													
PSO2	Ability to learn emerging computing paradigms for research and innovation.													
CO – PO Affinity Map														
PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO														
CO 1	3											1	3	2
CO 2	3	2	2	2				2	2	2		2	3	2
CO 3	3	3	3	2	3			2	2	2		2	3	2
CO 4	3	3	3	2	3			2	2	2		2	3	2

3 – Strong, 2 Moderate, 1 -weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	Level 2	Level 3
50%	60%	50%	60%	70%

Class	Topics to be covered	Mode of Teaching	In-Class Activities	Out- Class Activities	CO Mapping	Reference
1(L)	Introduction to Course	Smartboard Presentation	-	-	-	-
2(L)	Embedded System Definition, Classification, Examples	Smartboard Presentation	Brainstorming	Introduction to Embedded Systems – using Microcontrollers like Arduino	CO1	T1
3(L)	Embedded System Characteristics	Smartboard Presentation	Brainstorming	Introduction to Embedded Systems – using Microcontrollers like Arduino	CO1	T1
4, 5(P)	Introduction to Laboratory Experiments	Practical	-	-	-	-
6(L)	Definition of Computer	Smartboard Presentation	Brainstorming	Refresh computer architecture concepts	CO1	T1
7(L)	Processor Architecture	Smartboard Presentation	Discussion	Refresh computer architecture concepts	CO1	T1
8(L)	Microprocessor, Microcontroller, Digital Signal Processors	Smartboard Presentation	Discussion	Refresh computer architecture concepts	CO1	T1
9, 10(P)	Embedded Development Tools	Practical	Demo	-	-	-
11(L)	ARM Architecture	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
12(L)	ARM Registers	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
13(L)	ARM Reset, Memory	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
14, 15(P)	Understanding ARM Technical Reference Manual	Practical	Demo	-	CO2	-
16(L)	ARM Operating Modes	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
17(L)	ARM Addressing modes and operands	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1

18(L)	ARM Memory Access and Logical Instructions	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
19, 20(P)	GPIO Configuration, LED Interfacing	Practical	Demo	-	CO2	-
21(L)	ARM Arithmetic and Shift Instructions	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
22(L)	Introduction to Thumb ISA	Smartboard Presentation	Discussion	Analyze the architectural difference between ARM7TDMI and ARM CM4.	CO1	T1
23(L)	Interrupt structure	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=W8J6aDbZY7w	CO3	T1
24, 25(P)	Switches Interfacing	Practical	Demo	-	CO2	-
26(L)	Capstone Project Review 1	-	-	-	CO4	-
27(L)	Capstone Project Review 1	-	-	-	CO4	-
28(L)	Capstone Project Review 1	-	-	-	CO4	-
29, 30(P)	Seven Segment Display Interfacing	Practical	Demo	-	CO2	-
Mid Term Exam						
31(L)	Capstone Project Review 1	-	-	-	CO4	-
32(L)	Vector interrupt table	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=W8J6aDbZY7w	CO3	T1
33(L)	Interrupt service routines	Smartboard Presentation	Discussion		CO3	T1
34, 35(P)	LCD Interfacing	Practical	Demo	-	CO2	-
36(L)	Edge triggered Interrupts	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=NoBGeTqtj3g	CO3	T1

37(L)	Timer Periodic Interrupts	Smartboard Presentation	Discussion		CO3	T1
38(L)	Nested Vectored Interrupt Controller	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=x7TY-TkWEB0	CO3	T1
39, 40(P)	DAC Interfacing	Practical	Demo	-	CO2	-
41(L)	SysTick timer	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=x7TY-TkWEB0	CO3	T1
42(L)	Synchronous and asynchronous Communication	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=BR8aNAJXbhM	CO3	T1
43(L)	Capstone Project Review 2	-	-	-	CO4	-
44, 45(P)	ADC Interfacing	Practical	Demo	-	CO2	-
46(L)	Capstone Project Review 2	-	-	-	CO4	-
47(L)	Capstone Project Review 2	-	-	-	CO4	-
48(L)	Capstone Project Review 2	-	-	-	CO4	-
49, 50(P)	PWM Generation	Practical	Demo	-	CO2	-
51(L)	I2C-specification	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=CAvawEcxoPU	CO3	T1
52(L)	I2C Configuration	Smartboard Presentation	Discussion		CO3	T1
53(L)	SPI-specification	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=0nVNwozXsIc	CO3	T1
54, 55(P)	Real-Time Clock Interfacing	Practical	Demo	-	CO2	-

Quiz 1

56(L)	SPI Configuration	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=0nVNwozXslc	CO3	T1
57(L)	Moore Machines	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=kb-Ww8HaHuE	CO3	T1
58(L)	Mealey Machines	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=kb-Ww8HaHuE	CO3	T1
59, 60(P)	Timer Programming	Practical	Demo	-	CO3	-
61(L)	Performance Measurements	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=kb-Ww8HaHuE	CO3	T1
62(L)	Testing and Debugging of embedded applications	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=vEErk6-aanC	CO3	T1
63(L)	Capstone Project Review 3	-	-	-	CO4	-
64, 65(P)	Interrupt Programming	Practical	Demo	-	CO3	-
66(L)	Capstone Project Review 3	-	-	-	CO4	-
67(L)	Capstone Project Review 3	-	-	-	CO4	-
68(L)	Capstone Project Review 3	-	-	-	CO4	-
69, 70(P)	UART Programming	Practical	Demo	-	CO3	-
Quiz 2						
71(L)	Capstone Project Review 4	-	-	-	CO4	-
72(L)	Capstone Project Review 4	-	-	-	CO4	-
73(L)	Capstone Project Review 4	-	-	-	CO4	-

74, 75(P)	Capstone Project Review 4	-	-	-	CO4	-
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S.No.	List of Experiments	CO Mapping
1	(Teacher-led Activity – Not for Evaluation) Laboratory Experiments: Discussion on various possible application scope of embedded systems and development tools involved.	-
2	(Teacher-led Activity – Not for Evaluation) Embedded Development Tools Demo: Demonstrate the steps and tools involved in writing an embedded C program for ARM architecture, saving it, compiling it, debugging it and emulating it on a development computer system.	-
3	(Teacher-led Activity – Not for Evaluation) Understanding ARM Technical Reference Manual: To study the architecture and understand from Embedded C perspective.	CO2
4	GPIO Configuration, LED Interfacing: Write an embedded C program for ARM architecture to interface an LED. Debug and verify the program using IDE on a development computer system.	CO2
5	Switches Interfacing: Write an embedded C program for ARM architecture to interface switches. Debug and verify the program using IDE on a development computer system.	CO2
6	Seven Segment Display Interfacing: Write an embedded C program for ARM architecture to interface Seven Segment Display. Debug and verify the program using IDE on a development computer system.	CO2
7	LCD Interfacing: Write an embedded C program for ARM architecture to interface LCD. Debug and verify the program using IDE on a development computer system.	CO2
8	DAC Module: Write an embedded C program for ARM architecture to drive DAC module. Debug and verify the program using IDE on a development computer system.	CO2
9	ADC Module: Write an embedded C program for ARM architecture to drive ADC module. Debug and verify the program using IDE on a development computer system.	CO2
10	PWM Module: Write an embedded C program for ARM architecture to drive PWM module. Debug and verify the program using IDE on a development computer system.	CO2
11	Real-Time Clock Interfacing: Write an embedded C program for ARM architecture to interface Real-Time Clock. Debug and verify the program using IDE on a development computer system.	CO2
12	Timer Programming: Write an embedded C program for ARM architecture to drive Timer module. Debug and verify the program using IDE on a development computer system.	CO3
13	Interrupt Programming: Write an embedded C program for ARM architecture to drive Interrupt module. Debug and verify the program using IDE on a development computer system.	CO3

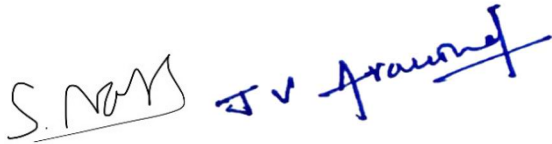
14	UART Programming: Write an embedded C program for ARM architecture to drive UART module. Debug and verify the program using IDE on a development computer system.	CO3
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Evaluation rubrics:

Capstone Project Evaluation (100 Marks)			
Evaluation Component	Excellent (100%)	Good (60%)	Fair (40%)
Understanding & Analysis of the Application, towards embedded system Development Feasibility (30 Marks)	The team has exhibited the in depth understanding and thoroughly analysed the problem (30x1=30 Marks)	The team has exhibited average understanding and incompletely analysed the problem (30x0.6=18 Marks)	The team has exhibited insufficient understanding and insufficiently analysed the problem (30x0.4=12 Marks)
Arriving on the specifications, Resource Requirements and Design Methodology (30 Marks)	The team has arrived on the commendable specs and design methodology (30x1=30 Marks)	The team has just arrived on the specs and just designed the system (30x0.6=18 Marks)	The team has arrived on the incomplete specs and insufficiently designed the system (30x0.4=12 Marks)
Generate & Executable the code, Run the application and Conduct the System & Acceptance Testing (30 Marks)	The team has generated & executed the code successfully . Testing is completed and Acceptance Testing is Cleared (30x1=30 Marks)	The team has generated & executed the code but with errors . Testing is incomplete and System Testing is in-progress (30x0.6=18 Marks)	The team has failed generate & execute the code . Testing is not done or Unit Testing is in-progress (30x0.4=12 Marks)
Teamwork, Presentation & Interaction (10 Marks)	The Teamwork was demonstrated and they presented the content satisfactorily (10x1=10 Marks)	The Teamwork was not convincing and they presented the content insufficiently (10x0.6=06 Marks)	The Teamwork was not at all demonstrated and their presentation is unsatisfactory (10x0.4=04 Marks)

Attendance Requirement:

- (i) All students must maintain a minimum of 75% attendance in each course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.
- (ii) Students who fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams. Requests for exceptions will not be considered under any circumstances.
- (iii) Attendance will be calculated up to three days prior to the commencement of the mid-term and end-semester exams, as per the academic calendar.

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